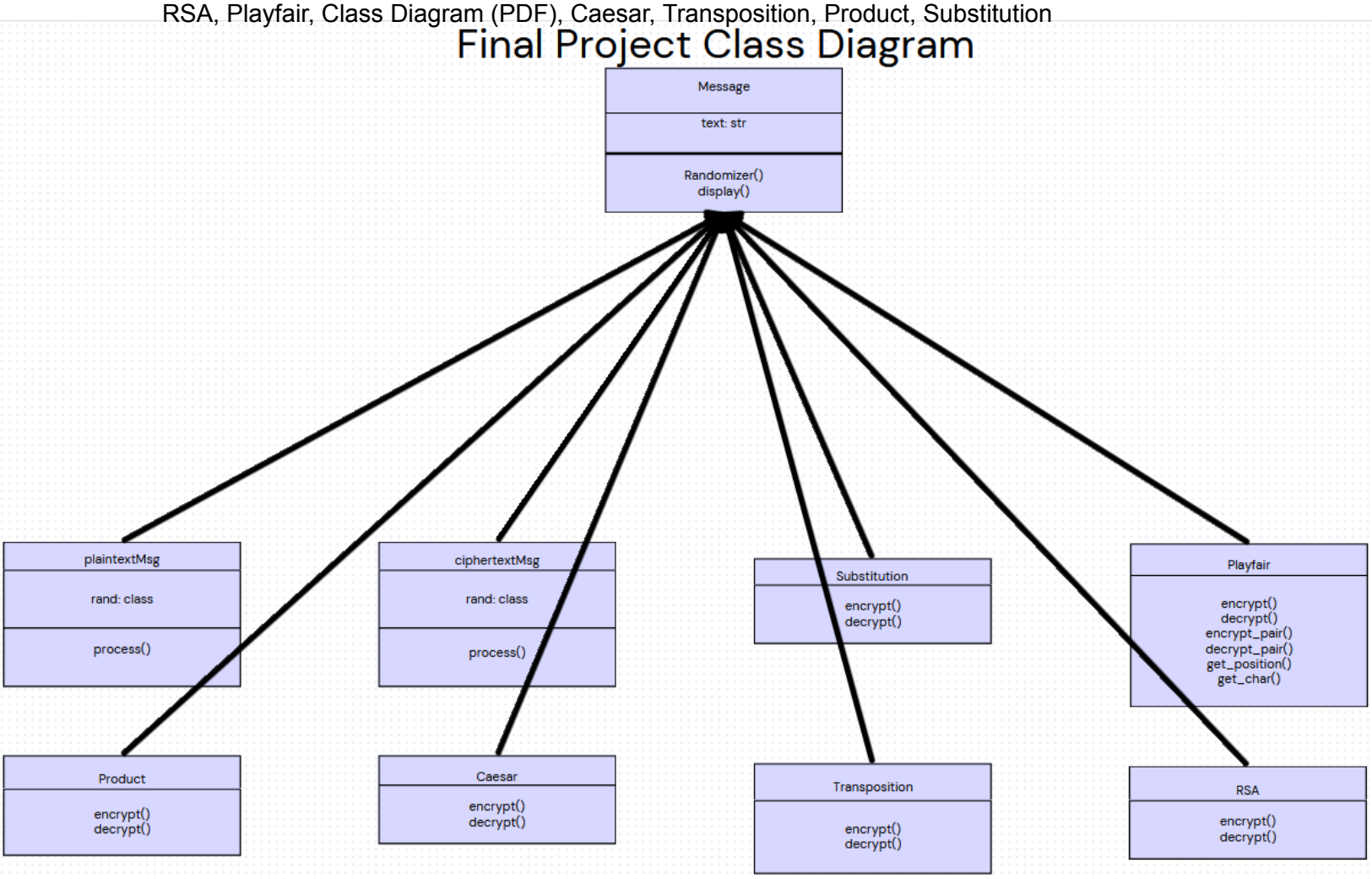
Final Project

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RSA, Playfair, Class Diagram (PDF), Caesar, Transposition, Product, Substitution



I first tried to figure out how the program's shell would work. I figured out that I could use the random library from Python to select the type of cipher used randomly. The random function would randomly select a class I would put that class in a variable and then use that variable to activate it. I then transformed the plaintextMsg class into a class that uses that class variable and applies the encryption by initializing the encryption method from said class. I did the same with ciphertextMsg but instead with decryption. Writing the encryption classes themselves was a very difficult task. The easier ones were substitution and cipher, but Playfair and Transposition were the hardest to implement in code form. This whole core format for the code made it very hard to try and troubleshoot since changing one thing can mess everything up from the way I set things up. In this way, I also had to make sure all classes had the same parameters and all had to have to encrypt and decrypt methods or else the random choice would give an error. If I had more time I would work on Playfair more since even after countless troubleshooting they have some glitches. But most of the rest should be bug-free the main fix was having a static key instead of trying to randomly generate one. That was causing more problems than it needed to. Overall this was a fun project that I would do again.